# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-J2-EN · v2.1 · 2026-11-01 |
| Package | star-j2 |
| Version | 0.1.0 |
| Capability | star.orbit.j2-secular |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_j2 |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_j2-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
| --- | --- |
| star_j2-0.1.0-py3-none-any.whl | b052661a1cb2d49516efc3362846f5dbafac36adb4a35d5f005d5e490a455dfe |
| star_j2.py | e97f9628cbc4ebb6cbfcf5c74143e1bc5a7dce2ba68e3192b2a4c259b82d460f |
| star_j2-0.1.0-source.zip | ea91eb8f3e1645a617afd6baef22dde1dae9dfb8454bddf1667abd6e6b30f037 |

# 5. What it does, and its limits

First-order secular effect of J2 on mean elements: node regression, perigee rotation, mean-anomaly rate, nodal period, sun-synchronous inclination; measured within 0.4 % of Orekit's Eckstein-Hechler theory and of a numerical integration.

Mean rates only: no short-period terms, no J2-squared, J3 or higher harmonics, no drag, no third bodies. Not a propagator. For mission design at the 0.5 % level; a station-keeping budget needs a full theory or an integration.

- `raan_rate_deg_day`, `argp_rate_deg_day`, `mean_anomaly_rate_deg_day`, `nodal_period_s` for mean elements (a in km, e, inclination in degrees), EGM96/WGS-84 constants by default, overridable by keyword.
- `sun_synchronous_inclination_deg(a_km, e=0)`: the inclination whose node advances 360 degrees per tropical year; `CRITICAL_INCLINATION_DEG` = 63.4349...: the perigee does not rotate.
- Accuracy, measured on 51 orbits: node rate within 0.31 % of Orekit's Eckstein-Hechler theory (J2 only, mean elements) and within 0.38 % of a numerical integration of the equations of motion with J2; perigee rate within 0.013 deg/day of the numerical integration. The formulas are first order in J2: differences of this size are the model, not a rounding error.
- Every function returns a finite float or raises `ValueError`: non-numeric, boolean, NaN or infinite input; a <= 0; e outside [0, 1); inclination outside [0, 180]; perigee inside the Earth; non-positive mu, re or j2; an orbit too high to be sun-synchronous (above about 12 350 km semi-major axis for a circular orbit).

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_j2_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-j2}: What the Earth's oblateness does to an orbit, on average},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_j2/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts